or replacement parts on the International Space Station. Groups such as Made in Space are developing tools that can directly manufacture in space – in zero gravity conditions – as and when needed.

- **Defense:** Much of the machinery used in the military is complex and produced in relatively low volumes. At the same time, they must be strong, durable and reliable. 3D printing technology has already been used to create parts that can survive the rigour of a battlefield, and very importantly, at much cheaper prices. Another problem with military equipment is relying on the limited amount of spares available – and with lives at risk, running out of required equipment can be disastrous. In the future, therefore, it may be possible for the military to print replacement parts on the spot instead of trying to rely on the inventory that they carry. Another – completely parallel – usage of 3D printing in the military would be to create topographical models on the spot to provide better understanding and intelligence to the military personnel. Regularly updated 3D models of the landscape would help soldiers in easily visualising, and hence responding better to a situation. This was also used recently by the U.S. Army to help respond to Hurricane Katrina.

3D printing in outer space

- **Automotive:** The automotive sector has already started using 3D printing to make complex parts. Major manufacturers such as GM, Land Rover, Bentley, GM and Audi have been 3D printing auto parts for a few years now. High end, specialty cars with relatively small production runs will particularly benefit. Engineers at BMW have used 3D printing to create lighter versions of their assembly tools to improve worker productivity. Also, 3D printing will take the marketing scenario of automobiles to a whole new, more visually detailed, dimension. Imagine showing a full scale 3D model instead of a CAD drawing. There's a lot to look forward to in the automotive sector – in fact, the world's first printed car, called